



**DGLRM**

# From Bed Rest to Mars:

## Development of a Literature-Derived Machine Learning Framework for Predicting Lower-Limb Muscle Atrophy in Spaceflight Analogues

Niloufar Ahmadymarzdashty<sup>1</sup>, Milad Bahari Qaragoz<sup>2</sup>

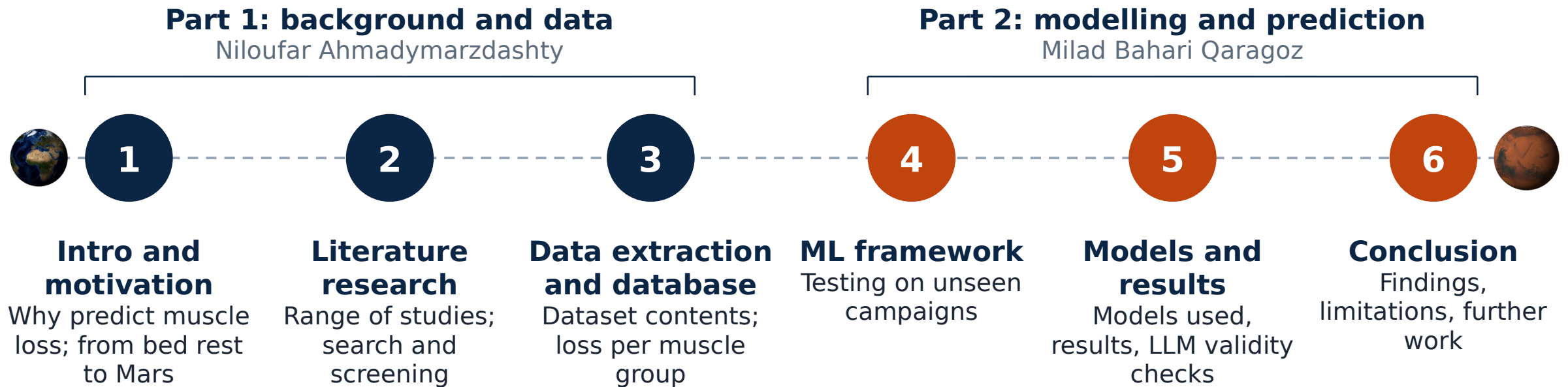
<sup>1</sup>Independent researcher

<sup>2</sup> Master's candidate, Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

**64. Jahrestagung der DGLRM**



# Agenda



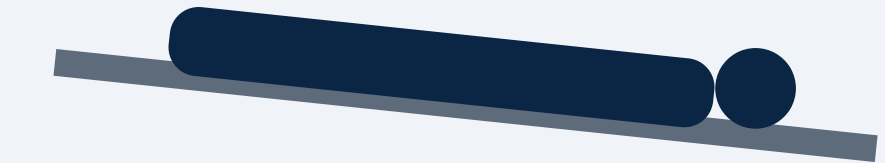
*The marker in the bottom-right corner of each slide shows where we are.*

# Motivation: predicting lower-limb muscle atrophy

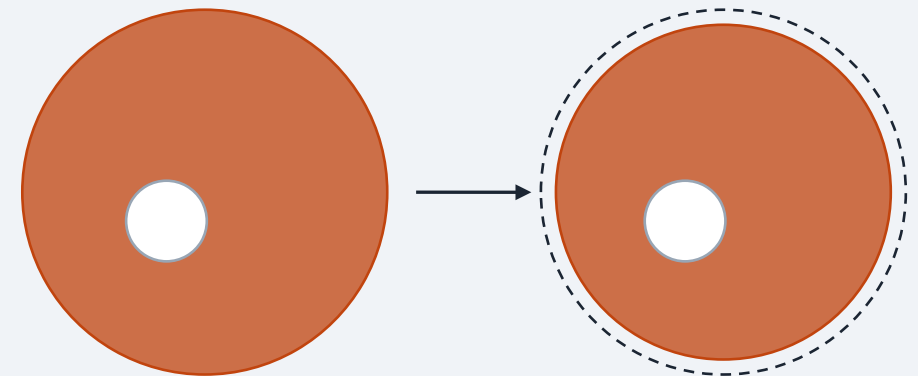
## Why is this important?

- Prevention: identify risk early and leave time to intervene
- Countermeasures: match exercise and nutrition to the predicted risk
- Vulnerable muscles: show which muscles need closer monitoring
- Function: less muscle loss helps preserve strength and power
- Mission planning: estimate muscle loss on long missions with limited exercise equipment

**Figure 1. Unloading shrinks leg muscles**



*Weeks to months in bed; legs carry no body weight*



**Before bed rest**

**After unloading**

*Schematic calf cross-section; dashed line = size before unloading.*

# From bed rest to Mars: analogue and application

## Bed rest: the ground analogue

- Spaceflight data on muscle are scarce
- Head-down bed rest unloads the legs under controlled conditions

## Mars: the long-term application

- No early return in an emergency: the crew depends on its own fitness

Figure 2. Moon mission vs Mars mission



Figure 3. From bed-rest data to crew readiness

Dashed: future use, which needs validation with spaceflight data

## Range of unloading durations and studies

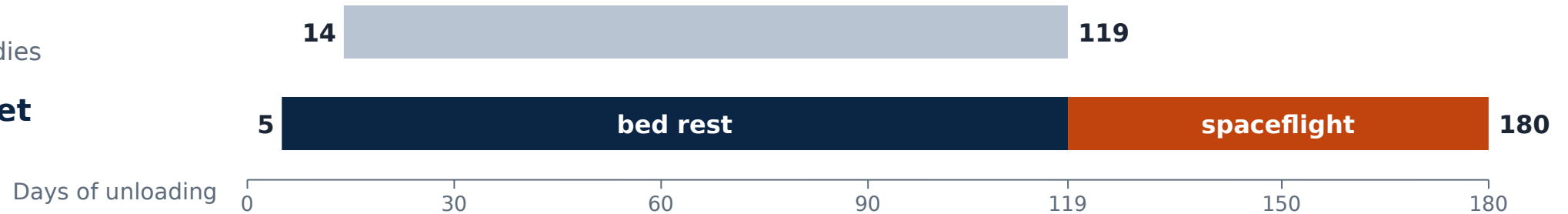
### Figure 4. Unloading range in the abstract and in the final dataset

## Abstract

## 15 bed-rest studies

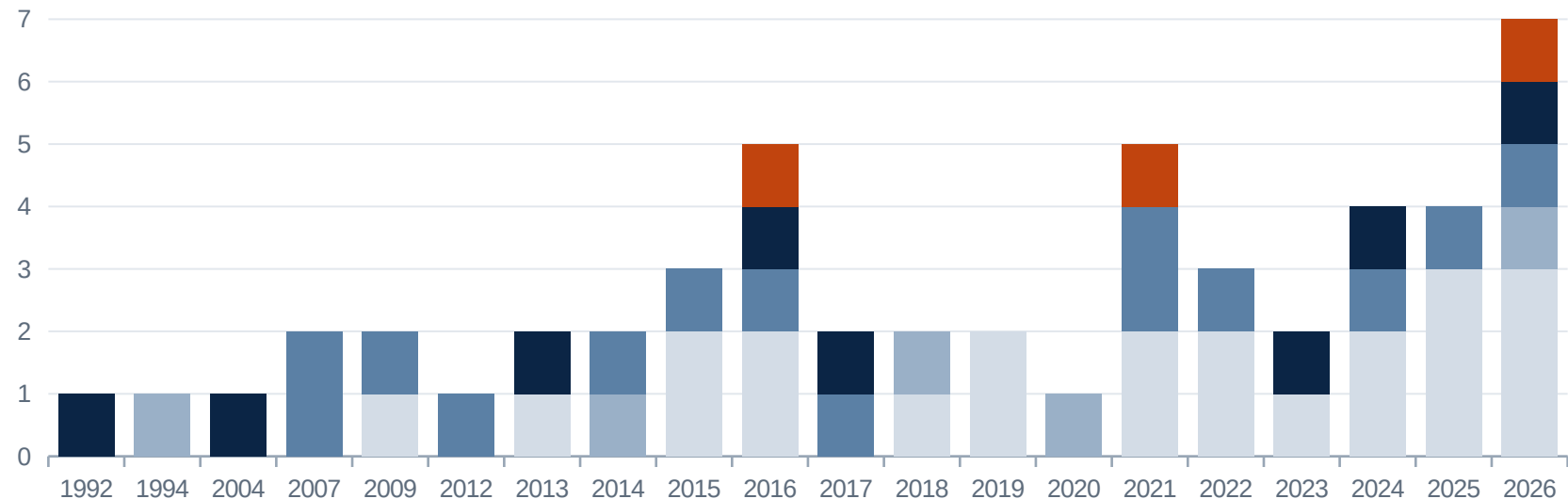
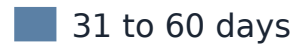
## Final dataset

52 studies



### Figure 5. Studies published per year, by length of unloading

*Years without studies omitted*



# Literature search and study selection



Figure 6. Flow of records, in the style of PRISMA 2020 (the standard for reporting systematic reviews).

# Dataset composition

Table 1. Final dataset at a glance

| Feature                          | n         |
|----------------------------------|-----------|
| Muscles or muscle groups         | 51        |
| Rows: control / countermeasure   | 470 / 272 |
| Rows: during bed rest / recovery | 478 / 264 |

Table 2. Imaging method and size measure

| Method     | n   | Size measure         | n   |
|------------|-----|----------------------|-----|
| MRI        | 655 | Volume               | 594 |
| DXA        | 37  | Cross-sectional area | 100 |
| CT         | 30  | Lean mass            | 37  |
| Ultrasound | 20  | Thickness            | 11  |
| Total      | 742 | Total                | 742 |

**742 observations are not 742 independent participants:** one campaign gives rows for several muscles, groups and scan days.

# Muscle atrophy by muscle group

Change in muscle size

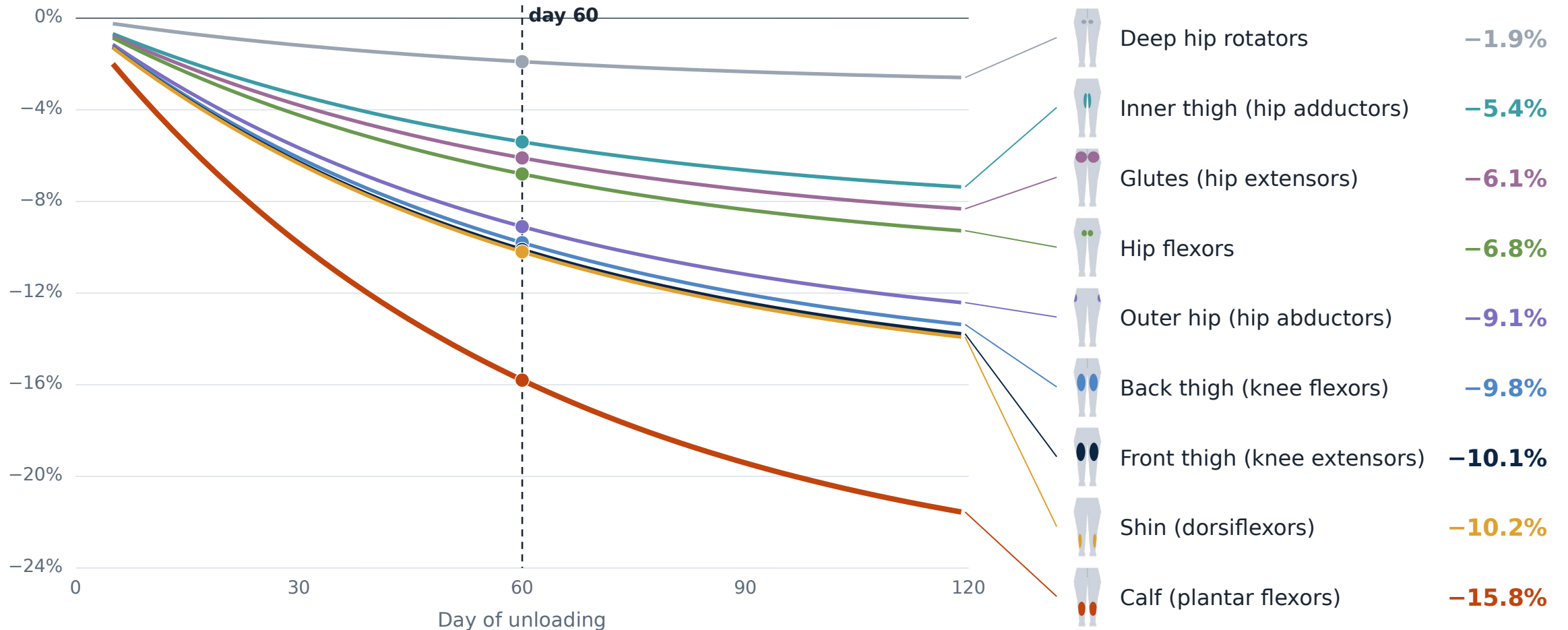
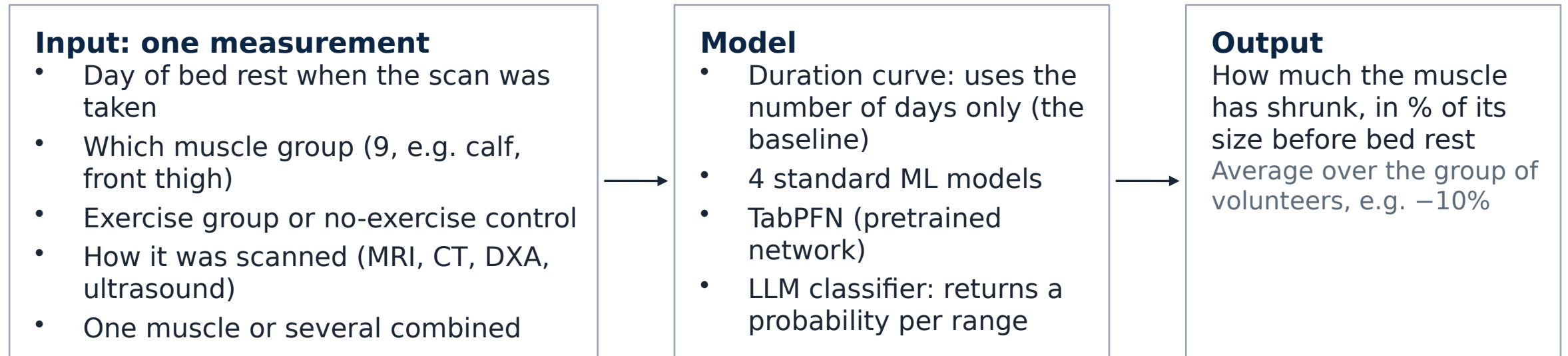


Figure 7. Change in muscle size by muscle group: one shared time course ( $\tau = 60$  days) scaled to each group's day-60 estimate; control groups, days 5 to 119.

# Evaluation: leave-one-campaign-out validation



**Figure 8. Testing: hold out one campaign, train on the rest, repeat 32 times**



Grey: 31 training campaigns. **Orange: the held-out test campaign.** Each campaign is held out once.

- Why campaigns, not measurements: the 346 rows used for modelling come from 32 of the 36 campaigns, and rows of one campaign share volunteers.
- Score: average error in percentage points (pp), each campaign counted once. Predicted -10%, measured -13%: error 3 pp.

# LLM-based probabilistic classifier: input and output

## Request (one per measurement)

### participants

men only, mean age 35, n = 11

### protocol

head-down tilt bed rest, 21 days, control group

### measurement

vastus lateralis, MRI cross-sectional area

### target

day 21 of bed rest

### typical\_curve\_other\_campaigns

duration curve fitted on the other 31 campaigns

### observations\_other\_campaigns

rows from the other 31 campaigns, most similar first, up to about 26,000 tokens

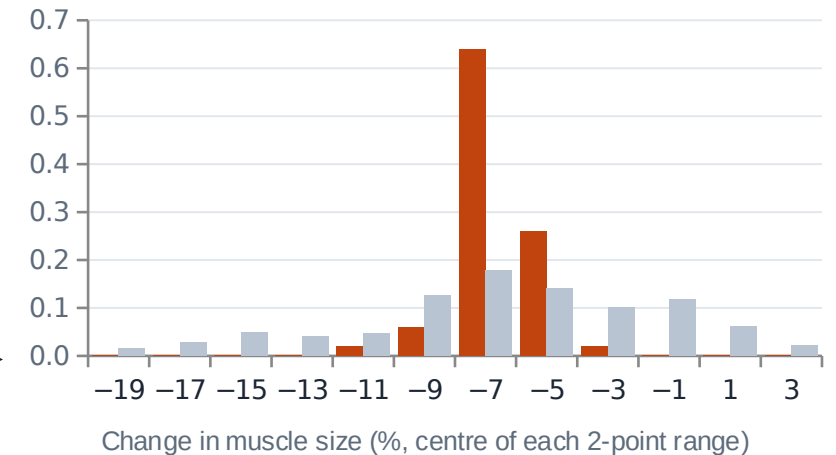
### question

Which range will the change fall in? 19 options: below -30%, 2-point ranges up to +4%, +4% and above



*Not a chat model: it is built for classification and returns only a probability for each listed option, never free text.*

Figure 9. Probability per range



■ LLM ■ Duration curve

- Our code turns the probabilities into one prediction, their weighted average: -6.6%
- Measured: -9.4%, an error of 2.8 pp. This is a typical row: half of the LLM's errors are smaller.

**Never sent (checked on every request):** the held-out campaign's measurements, and any paper, author or campaign name. All 430 answers are saved, so every result can be rebuilt without calling the model again.

# Model comparison on held-out campaigns

Figure 10. Error, pp (lower = better)

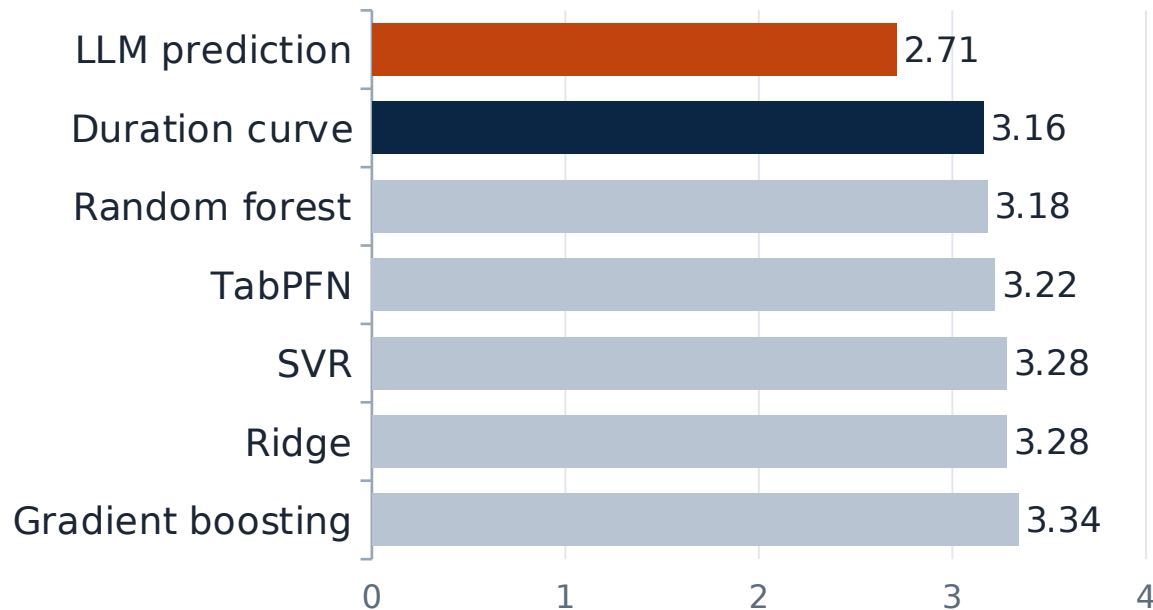
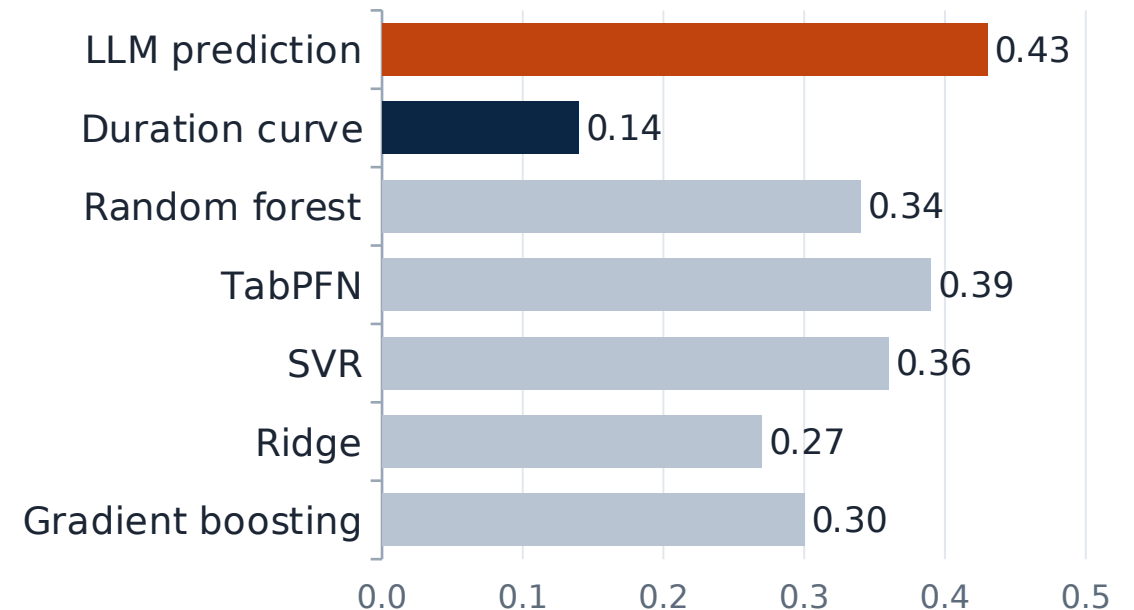


Figure 11.  $R^2$  (higher = better)



- ML models explain more of the differences ( $R^2$  0.27 to 0.39), but none has lower error than the curve.
- LLM vs curve, scored the same way: 2.71 vs 3.13 pp, so 0.42 pp smaller; closer in 20 of 32 campaigns.
- 95% confidence interval of that difference: 0.13 to 0.73 pp. The whole range is above zero, so it is unlikely to be luck.

*The curve shows 3.13 pp here, not 3.16, because the comparison scores it exactly as it scores the LLM.*

# Validity checks of the LLM prediction

*Gain = how much smaller the LLM's error is than the curve's, in pp (full result: 0.42). Each check and its pass rule were fixed before running it.*

## Does it recognise a known campaign?

- Everything identifying a campaign removed: gain still 0.30 pp, interval above zero
- Asked to choose the campaign's name from a list of options: right 19% of the time, where guessing gives 8%

## Does it use the other campaigns' data?

- Their values shuffled: worse than the curve (gain  $-1.25$  pp)
- None of their data given: error 5.85 vs 3.13 pp

## Is it sensitive to presentation?

- Other campaigns' rows in a different order: error changes 3%
- Answer ranges moved by 1 point: error changes 1%

## Is it reproducible?

- 20 identical requests resent: predictions differ by 0.17 pp on average, well below the 0.42 pp gain

## Limitations of this result

- Added after the ML result was known; not in the original plan
- Overconfident ranges: its 80% ranges hold the measured value 62% of the time, so we report only its single-number prediction
- Given a campaign's own earlier scans, it does not use them

# Conclusions, limitations and further work

## Conclusions

- ML models do not beat a duration-only curve (3.18 vs 3.16 pp)
- The LLM classifier lowers the error to 2.71 pp (curve 3.13 pp)

## Limitations

- 32 independent campaigns, mostly young men
- No bed-rest data beyond day 119
- Group averages; bed rest is not spaceflight

---

## Further work

- Link the muscles: predict one muscle's loss from another's, and weight them into an estimate of strength loss
- A risk-assessment system based on a person's characteristics, such as age, sex and fitness



**DGLRM**

# Thank you for listening

**Questions are welcome**

**From Bed Rest to Mars | 64. Jahrestagung der DGLRM**



# Backup: dataset coverage and limitations

## Did you miss studies?

- Four sources searched from 2013 on, plus 10 older or open-data studies (sources and queries on the next slide)
- Not covered: Embase (no access); 1,023 records not yet screened (screening backup)

## One campaign dominates the data

- MEDES 90-day: 40% of all rows, a quarter of the modelling data
- Scoring gives each campaign one vote; the re-run without it is still to do

## Can MRI, CT, DXA and ultrasound be mixed?

- Each method gets its own term: DXA and ultrasound thickness show 5 to 6 pp less loss than CT; MRI volume is within 0.5 pp
- The MRI-only re-run is still to do

## How reliable is the extraction?

- Every value is traced to its paper and page, table or figure
- 674 of 742 rows high confidence; 19 read off figures; no independent second extraction

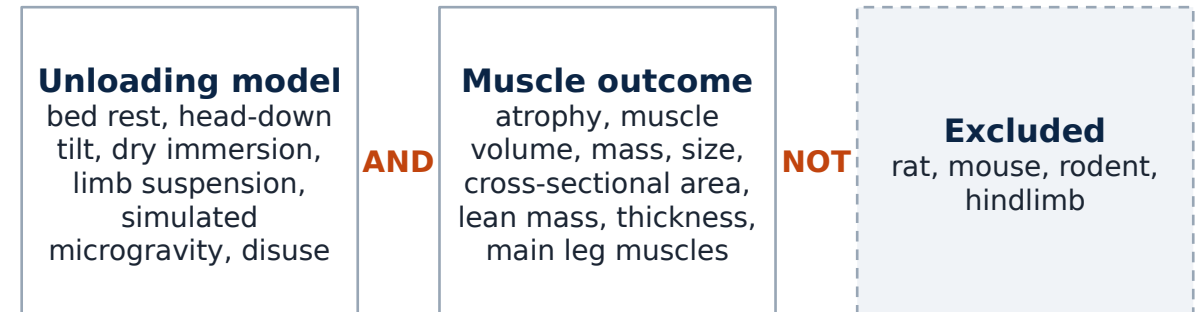
# Backup: literature search sources and query

**Table 3. Sources (searched 4 September 2026)**

| Source                             | Records      |
|------------------------------------|--------------|
| PubMed                             | 1,412        |
| Scopus                             | 1,757        |
| Web of Science Core Collection     | 1,876        |
| NASA Technical Reports Server      | 686          |
| Earlier studies and NASA open data | 10           |
| <b>Total</b>                       | <b>5,741</b> |

- Journal databases: publications from 2013 onwards
- Older work enters through studies we held before the search (9) and one NASA open-data campaign
- NASA reports: five short keyword queries, e.g. "bed rest" "muscle volume"
- Not searched: Embase (no institutional access), Cochrane CENTRAL, trial registries, citation chasing

**Figure 12. Query: two required blocks**



**Web of Science string, verbatim**

```
TS=((("bed rest" OR bedrest OR "head-down tilt" OR "head down bed rest" OR HDBR  
OR antiorthostatic OR hypokinesia OR hypodynamia OR "dry immersion"  
OR "limb suspension" OR ULLS OR "simulated microgravity"  
OR "microgravity analog*" OR "spaceflight analog*" OR "mechanical unloading"  
OR "muscle unloading" OR disuse)  
AND  
(atroph* OR "muscle volume" OR "muscle mass" OR "muscle size"  
OR "cross-sectional area" OR PCSA OR "lean mass" OR "muscle thickness"  
OR "muscle wasting" OR deconditioning OR soleus OR gastrocnemius  
OR "triceps surae" OR quadriceps OR "vastus lateralis" OR "knee extensor"))  
NOT TS=(rat OR rats OR mice OR mouse OR rodent OR hindlimb OR "hind limb")
```

# Backup: screening and study selection

## **3,600 screened, 2,493 excluded: where are the other 1,107?**

- 84 went on to full text: 74 from the search and 10 from other sources
- 1,023 were not screened in the time available: the rest of the "maybe" set and 80 records waiting for a full text
- They are reported as unscreened, not as excluded

## **84 full texts but 52 studies: why so few?**

- 19 give muscle results only as charts, without a baseline value
- 4 full texts not available (one conference abstract still used); 11 excluded at full text

## **Who screened, and was it done twice?**

- Eligibility criteria were written before screening began
- A rule-based triage sorted the records; one person then read the priority set and the top of the "maybe" set
- No second screener

## **Why search only from 2013?**

- The search was time-boxed to one week
- Older work enters through 9 studies we held before the search; 6 of their campaigns would otherwise be missing
- Both known modelling papers from after 2013 were found by all three journal databases

# Backup: dataset expansion and composition

## The abstract had 15 studies. Did adding 37 change the analysis after seeing data?

- Eligibility criteria were fixed before screening; the dataset was frozen (14 and 19 September) before the final models
- The curve shape was chosen by a rule declared before fitting

## Mostly young men: does this apply to women and older crews?

- Rows: men only 567, mixed 113, women only 40 (one campaign, WISE-2005)
- Healthy young 692, older 42: the results describe young men

## Slide 5 shows 180 days, but the model stops at day 119. Why?

- The 14 spaceflight rows were measured after landing, so they leave the model with the 264 recovery rows
- 8 of the 14 are one back muscle (lumbar multifidus)

## What exactly is a row's percentage change?

- Follow-up vs the same group's own baseline; negative means loss
- Recomputed from the printed values; where a paper prints the mean of individual changes, that value is kept and labelled
- Rows weighted by number of participants: only 8 of 32 campaigns report the spread of the change

# Backup: certainty of the muscle-group curves

## Is each line fitted to its own muscle?

- No. The day-60 values are fitted per muscle group (named muscles, 25 campaigns); the lines share one time course ( $\tau = 60$  days)
- Separate shapes cannot be estimated: six of nine groups rest on four campaigns or fewer

## How certain are the day-60 values?

- Calf  $-15.8\%$  (95% CI  $-18.9$  to  $-12.7$ ); front thigh  $-10.1\%$  ( $-13.6$  to  $-6.6$ ); hip rotators  $-1.9\%$  ( $-3.9$  to  $0.2$ )
- Calf vs shin, the ankle's opposing pair: 5.6 pp more loss (2.1 to 9.1),  $p = 0.003$
- Hip rotators and outer hip: one campaign each

## Why a saturating curve?

- Logarithmic, saturating and spline curves fit equally well (within 1.3 AIC points): the loss slows, but the data cannot say along which curve
- By day 119 the curve reaches 86% of its plateau,  $-17.3\%$  (95% CI  $-20.9$  to  $-13.7$ )

## Is the calf result an artefact of how often it is scanned?

- The order holds after adjusting for duration, countermeasure group and imaging method
- Calf: 78 rows from 13 campaigns, the second best-covered group after the front thigh (86 rows, 22 campaigns)

# Backup: significance and relevance of the LLM gain

## Is 0.42 pp worth anything?

- It is 13% lower error than the curve (3.13 to 2.71 pp)
- Small; we present it as a secondary finding

## 20 of 32 campaigns is close to a coin flip

- Counting wins only:  $p = 0.22$  (sign test)\*
- Using the size of each win: mean gain 0.42 pp, 95% CI 0.13 to 0.73

## Does it rest on a few campaigns?

- 3 long MRI campaigns (NASA SPRINT, Berlin BBR2-2, MEDES LTBR) give 56% of the gain; without them 0.20 pp, CI  $-0.01$  to  $0.43^*$
- Only SPRINT is recognised, and with campaign details removed the LLM still wins there (3.79 vs 5.92 pp)

## Did it read these papers in training? Was it added after the fact?

- Training exposure cannot be ruled out by any test; hiding campaign details and the naming test are the strongest checks available
- Added after the ML result, but every check and target was committed before the first answer

*\* Our calculation from the saved LLM predictions; not in the report.*

# Backup: ML results and fairness of the comparison

## Did the ML models get a fair chance?

- Tuned inside each training fold, grouped by campaign; TabPFN needs no tuning
- All models saw the same inputs, folds and scoring

## Why not add age, sex or countermeasure type?

- 32 campaigns leave room for about 3 campaign-level inputs (about 10 campaigns per input)
- Age and sex are fixed within a campaign (347 of 425 rows men-only), so they act as a campaign label; 126 countermeasure rows spread over 9 types

## Is the duration curve a straw man?

- Ridge regression uses days, muscle and every other input: 3.28 vs 3.16 pp
- Knowing the muscle raises  $R^2$  but not the error on a new campaign

## Would a nearest-neighbour method match the LLM?

- Not tested. If the gain comes from picking similar rows, this is the next control to run

# Backup: applicability to Mars mission planning

## A Mars transit is about 180 days; the data stop at 119

- Only one campaign reaches 119 days
- A 180-day estimate is planned from the duration curve only, with an uncertainty range, labelled as an assumption

## Which countermeasure works? What about individuals?

- Countermeasure groups lose 3.7 pp less (95% CI 1.5 to 5.9), all types pooled
- Every row is a group average; nothing about individual risk

## Bed rest is not spaceflight

- Only 14 spaceflight rows, 8 of them one back muscle: too few to test transfer

## Can anyone reproduce an LLM result?

- Fixed model version; all 430 answers saved, so every number rebuilds without the model
- Identical requests differ by 0.17 pp on average